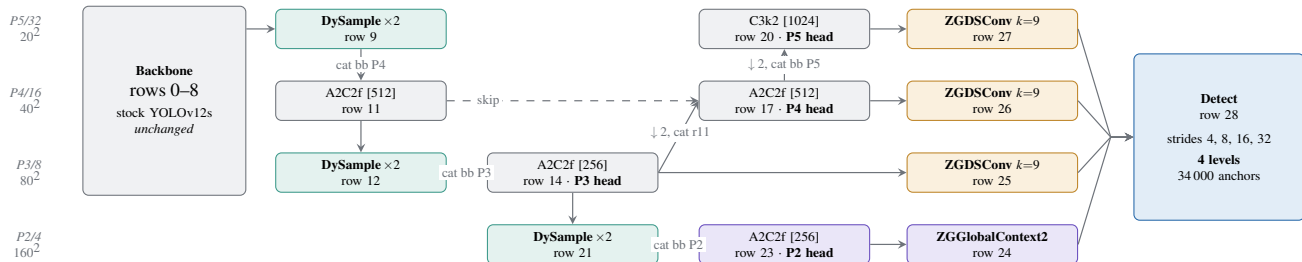
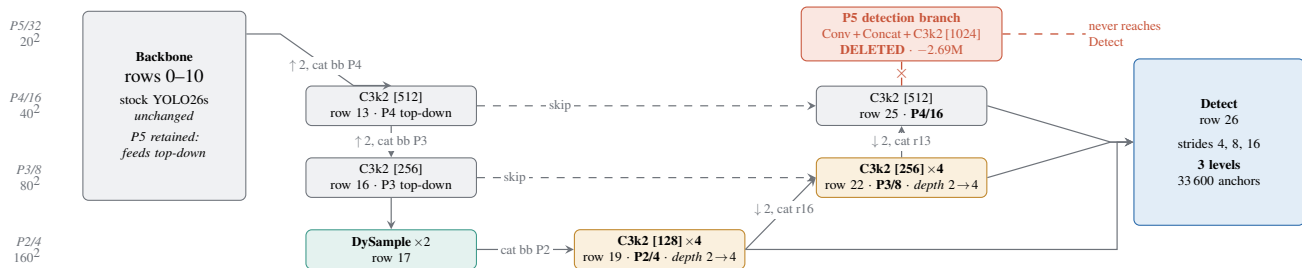


(a) YOLOv12s + 1s_shift - the stride-4 branch is *appended*; stride 32 is kept $\Rightarrow$ four detection heads



(b) YOLO26s + p2dys-p234rich - stride 4 is *inserted into the PAN*; stride 32 is *removed* $\Rightarrow$ three heads, -27.7% parameters



Legend

 stock block, unchanged	 stock operator replaced
 new branch	 capacity reallocated
 removed	 detection head

Anchor budget and cost at 640×640

level	map	anchors	
P2/4	160^2	25 600	+0.09M params but ≈ 315 MB activations
P3/8	80^2	6 400	
P4/16	40^2	1 600	81% of a large object's positives land here
P5/32	20^2	400	4.8% of the stock budget; 0.01 positives per small GT

Adding P2 multiplies the anchor count by four (8400 $\rightarrow$ 34000) for almost no parameters, which is why the architectural runs needed a smaller batch than the baseline and why a matched-batch control is reported. Removing P5 recovers only 400 anchors, so it is a parameter saving (-27.7%), not a memory one.